Energizing water through dynamic flow - Phase 1

Conclusive Report

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Summary

The aim of the study was to investigate the "state of energy" of a dynamic active water system consisting of four different Flowform designs (FF). The study was performed on two ceramic single vessels (Manawa vase and the Matatiki bowl), the three vessel Glonn vertical stack model and the 12 vessels vertical stack Greenhouse model. The results show that all four FFs transform less organized low-grade thermal energy obtained from the environment to high grade energy, which was used to make a purposeful work. The excited state of water was measured as an increase in pH and a decrease in redox potential and equally a decrease in conductivity, which correlated with an increase in fractal scaling as measured by detrended fluctuation analysis of thermal IR emission from surface water, with a concomitant decrease in water surface temperature in two of the four tested FFs, the Manawa vase and Greenhouse stack. The strong fractal power-law relationship revealed that fractal scaling geometry is part of coherent water ordering and that thermal IR flickering relates on a long-range correlation between water molecules, which is considered as manifestation of the underlaying quantum dynamics.

Background

The background is given in detail in proposal 2. The main phase 1 research question to be answered in the present study was if the combination of thermal IR imaging (TIRI) and detrended fluctuation analysis (DFA) can verify a change in the energy state measured as a coherent reorganization in water in either of the four FFs. A methodological change was introduced after approval of proposal 2. A pre-study on ordinary chemical parameters, i.e. pH, redox potential and conductivity revealed that the applied FFs could significantly affect basic water chemistry with a potential to differentiate the energetic outcome between FFs by conversion of low grade-energy towards a high-grade energy state of water. Accordingly, the three ordinary chemistry methods replaced the more complex differential electrochemical impedance spectroscopy (EIS) methodology.

Material and Methods

Spring water was obtained fresh for each test daily and circulated through FFs for two hours before sampling of 100 ml water for physical, chemical and TIRI analyses. Control water was equally conditioned at ambient temperature for two hours.

Flowform Designs are described in detail elsewhere (2) with their principal outlook shown in Figure 1. The water outflow flow rate was as follows; Manawa vase 5 lpm, Matatiki bowl 5.5 lpm. Greenhouse 2,5 lpm and Glonn 14 lpm.

Figure 1. *From left to right: Manawa, Matatiki, Greenhouse and Glonn designs.*

Submersible pumps; electrically powered with impellers. The two ceramics were each supplied by a SUNSUN HJ 731 pump with a magnetic rotational rod axle, driving wing impellers. The Green house was a Kockney KKYFP2400 pump and the Glonn was supported by an Elpump CT 3274 pump. All pumps were placed in the body water with approximately 15 cm water covering the upper part of the pump.

Redox potential (ORP), pH and conductivity were measured using an Inolab 740 multifunctional meter (WTW International, Stockholm, Sweden) according to our previous study on distilled water (1).

Thermal infrared imaging (TIRI) – Thermal IR frames (1 square cm) was obtained by a RAZ IR Thermal camera system (Sierra Pacific Innovation Corporation, US) from the top of two standing optical polystyrene cells containing 5.0 ml of control or experimental water (1). Six repeated image frames were made in sequence on all together 20 water samples collected from each FF (measured occasionally for 2 months) and saved for imagery analysis (temperature profiles of the pixel area and graphical representation of temperature data) with an IR Analyzer software.

Detrended Fluctuation analysis (DFA) determines the fractal scaling of a numeric temperature series data in the presence of possible trends without knowledge of their origin. The DFA algorithm permits detection of intrinsic self-similarity embedded in non-stationary temperature series. The change in DFA exponent was calculated on mean temperature data from thermal IR emission (TIRI analysis) pixels obtained on a 1.0 square cm water surface area (1). A DFA exponent (d) ranging from $0.5 \leq d \leq 1$ indicates long-range power-law correlations, growing stronger towards 1.0.

Water temperature in control water and in water obtained from the FFs at the end of equilibration time was measured by an electronic temperature sensor (Electronic digital thermometer, Traceable® Ultra, VWR, Sweden).

Time of the study – All together twenty measurements on sampled water were performed occasionally (one-day work at each occasion) on each of the four FFs from June 13 to August 8, 2018.

Solar geomagnetic activity (GMA) was followed during the experimental days (http://tesis.lebedev.ru/en/magnetic_storms.html). However, current solar GMA measured as solar radio flux, mean planetary A index and mean planetary Kp index were all normal or close to normal levels during study days and therefore without impact on fractal ordering of water flowing over FF surfaces.

Results

The mean results of the water temperature measured at the end of equilibration time of circulating FF water, the fractal scaling properties (DFA) and water chemistry (pH, redox potential (ORP), conductivity) are shown in Table 1.

Water temperature
The mean ± SD water temperature of control water was 20.5 ± 1.4 ºC. The water temperature of the two ceramics was non-significantly increased to 20.9 ± 1.6 ºC (Manawa vase) and 21.1 ± 1.7 ºC (Matatiki vase). A visual impression from the rhythmic movement of the water vortex form flowing in the bowl of the two ceramics gives a more confined flow-path shape of the Matatiki vase. The impact of serial multi-vessels (Greenhouse and Glonn stacks)

Table 1. Thermal infrared (IR) emission imaging, fractal scaling and water chemistry in spring water circulating in flowforms for two hours.

Flowform	Water Temperature		Fractal scaling			Water chemistry		
Sample[a]	Temperature (ºC)	Temp Diff (ºC)	TIRI Temp (ºC)[b]	TIRI Temp Diff (ºC)[b]	DFA[c]	pH[d]	ORP (mV)[d]	Conductivity (µS)[d]
Control water	20.5 ± 1.4		25.08 ± 1.59		0.823 ± 0.035	7.76 ± 0.09	139 ± 8.0	296 ± 2.9
Manawa vase	20.9 ± 1.6	0.35 ± 0,46	24.97 ± 1.60***	0.11 ± 0.10	0.871 ± 0.048***	8.51 ± 0.03***	123 ± 7.1***	290 ± 3.2***
Matatiki vase	21.1 ± 1.7	0,60 ± 0.83	25.28 ± 1.8		0.822 ± 0.052	8.52 ± 0.02***	123 ± 8.8***	290 ± 3.3***
Green House	21.3 ± 1.3***	0.78 ± 0.32	24.96 ± 1.8***	0.11 ± 0.08	0.864 ± 0.071***	8.47 ± 0.04***	124 ± 8.3***	292 ± 1.6***
Glonn	27.0 ± 1.7***	6.77 ± 0.76	25.05 ± 1.7		0.817 ± 0.066	8.58 ± 0.04*** [θ]	123 ± 5.7***	291 ± 1.9***

***P<0.001
[θ]P<0.001 Glonn compared to Matatiki Vase
[ϕ]P<0.001 Glonn compared to Green House
Control water is a local spring water
[a]Water strored in polystyrene cells. All values are an average of 20 water samples analyzed occationaly for 2 months on each Flowform.
[b]TIRI Temperature was calculated on numerical pixel data, each sample calculated on six repetitive image frames taken in sequence.
[c]DFA = Detrended Fluctuation Analysis calculated on serial sets of numerical temperature values according to pixel data.
[d]Redoxpotential (ORP), pH and conductivity were analyzed separted and independent of thermal TIRI measurements.

shifted the water flow towards a more complex movement giving a significant increase in water temperature. The former vessel stack with a less impingement gave a minor increase in water temperature (21.3 ± 1.3 ºC; P<0.001), while the thermal influence from the Glonn stack gave a huge increase in the water temperature up to 6.8 ºC (27.0 ± 1.7 ºC; P<0.001). A high flow rate and turbulent flow of water with a steady formation of a high concentration of air bubbles within the flowing water may contribute to the elevated temperature (3).

Fractal scaling and cooling of water IR temperature

The reduction in thermal IR emission and water surface temperature that differs from control water is shown in Figure 2. From the thermal IR control image (A), the structural ordering of control water demonstrates that the red colour zone (temperature decreases according to the scaling; blue-blue, red-red and dark red) along and close to the PET surface follows a subsiding circular low-density temperature gradient. The IR emission from the more distant regions in the centre radiates normally. The IR radiation from the FF water (B) close to the interfacial zone expands towards the centre and covers the entire water surface following a less density gradient towards the centre. The structural ordering and symmetry layer comprises the entire water surface.

A) B)

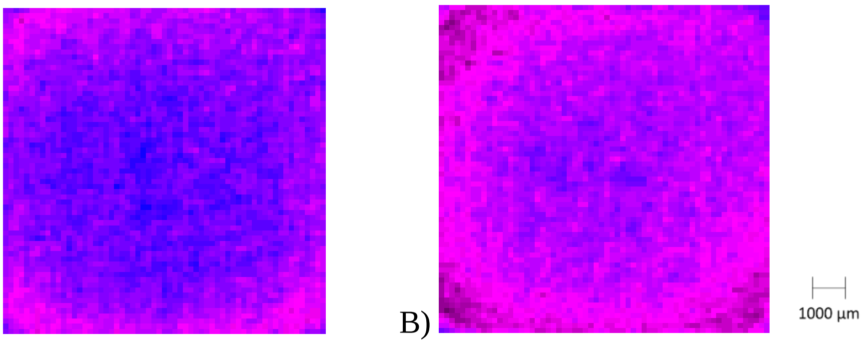

Figure 2. *Thermal IR emission at room temperature in control (A) and FF water (B) obtained from the Manawa vase stored in 5 ml plastic PET cells. The image of FF water depicts a higher state of structural organization of liquid water. The mean temperature was in average 0.3 ºC lower in experimental water. The temperature is decreasing according to the scaling; blue-blue/red-red and dark red colour.*

The results of DFA analysis (Table 1) revealed a clear power law relationship of the thermal IR emission in water obtained from the Manawa vase and Greenhouse stack. The DFA value increased significantly (P<0.001) from 0.823 ± 0.035 (control) to 0.871 ± 0.048 (Manawa vase). A similar irregular trend was obtained in water from the Greenhouse stack, 0.864 ± 0.071 (P<0.001). The DFA change indicates that the flickering of thermal IR emission is characterized by highly persistent long-distance-attraction due to non-thermal interaction between the molecules of water. In coherent water domains (CDs) (4,5) the thermal fluctuation aligns with and unifies the magnetic field oscillations of freely movable electrons, which leads to long-range oscillation and structural stabilization among coherent water clusters (1).

The characteristics of structural ordering of water molecules illustrated by the DFA change correlates with simultaneous decrease in water surface temperature. The IR emission becomes less and shifts towards higher wavelengths (red-shift). The water cools down where the ordered molecular structure of water from the Manawa vase generated a decrease in mean surface temperature of 0.11 ± 0.10 ($P<0.001$) and comparable to that of the Greenhouse stack, 0.11 ± 0.08 ($P<0.001$). Since the difference in the IR emission of the two FFs in relation to the control experiment equals a tuned decrease in surface temperature, the lower emission substantiates an intrinsic ordering of water molecules that occur due to the nonlocal characteristics of the flow-path of these two FF. At the descending temperature interfacial water tunes into an amorphous non-crystalline-like state with local and distant configurative physical properties (6,7).

Change in pH, ORP and conductivity

The significant change ($P<0.001$) in pH, ORP and conductivity of all four FFs (Table 1) aligns with the water state of CDs where the importance of long-range ordering with exceptional stability and feedback dynamics reflects the self-organizing ability with extended coherence of the inherent oscillations produced by the system itself (8,9). As compared to pH of the control, 7.76 ± 0.09, in the environment of the dynamic active water a significant rise was observed in all FFs ($P<0.001$), which equals 8.51 ± 0.03 (Manawa vase), 8.52 ± 0.02 (Matatiki vase), 8.47 ± 0.04 (Greenhouse stack) and 8.58 ± 0.04 (Glonn stack). The pH of the Glonn stack water was concomitantly significantly higher ($P>0.001$) than in the other three FFs. The exceptional high-water flow rate of the Glonn stack will evaporate gases rapidly and keep less carbon dioxide dissolved in water giving the higher pH of flowing water.

The change towards a reduction in ORP accompanied by the rise in pH of ordered FF water indicates self-regulative dynamics of circulating water aligned with long range oscillations and lowering of the intrinsic entropy, giving rise to structural ordering and allometric scaling behaviour, all of which are characteristic of a dissipative structure (10-12).

Accordingly, the significant change ($p<0.001$) and decrease in ORP substantiate the existence of stabilized CDs that enable a reservoir of quasi-free electrons, where the ORP was 139 ± 8.0 mV (control) and with lower values in water from all the FFs as follow, 123 ± 7.1 mV (Manawa vase), 123 ± 8.8 mV (Matatiki bowl), 124 ± 8.3 mV (Greenhouse stack) and 123 ± 5.7 mV (Glonn stack). The reduction in ORP and the concomitant rise in pH show an elevated stock of free electrons from an existing coherent state of liquid water.

The descending temperature of a coherent water state will dissolve less mineral salts and with a tendency to exclude minor particles as the coherent ordering of water increases and therefore the conductivity (EC) will subside (13). The EC was equally and significantly ($P<0.001$) decreased from baseline (296 ± 2.9 µS) in water from all four FFs and varied at its most from 290 ± 3.2 µS (Manawa vase) to 292 ± 1.6 µS (Greenhouse stack).

Discussion

In this study the impervious surface-designed flow-path of FF contribute to the natural low entropy vortical flow dynamics of water. The vortical flow refers to the electrical negative charge of high-grade energy water CDs moving in a loop enabled by the flow-path shape of FFs and producing circulating magnetic fields (4,5). According to the theory of dissipative systems' ability to self-regulate (12), it was implied that the exchange of energy and entropy with the environment relied on the fact that the absorption of external high entropy thermal energy (rise in water temperature) by the water vortical path was tuned into low entropy high-grade energy, which displaces the equilibrium towards a dominance of coherent water CDs. Like-wise, the CDs of FF-water were characterized by a distinctive configurative coherent aqueous state with fractal scaling low entropy relationship in radiative TIRI temperature fluctuations that maintained a self-organized structural adaptation irrespective of the water temperature (1). Consequently, the impact of an imposed disordering of water structure according to classical theory due to the rise in temperature of flowing water, is anticipated to local effects that extends no more than nanometers between molecular structures (13). The existence of CDs implies the presence of nonlocal ordering magnetic fields in ordered water. Since the ordered water is in a state of quantum coherence, it cannot decay thermally. Instead its lifespan extends for up to days or weeks and more (14,15). The thermal IR emission intensity from water is a function of temperature and structure (16). Because of the structural stability of coherent domains, the difference in emissivity therefore correlates with the considerably lower thermal IR emission compared to control water. The sustained and condensed temperature gradients with a consistently lowered magnitude in temperature that last for days are proof of a remarkable stability of the exited states in ordered water that agrees with the consistency in increased pH of succussed water (14).

When high-grade energy from the quantum field was captured by incident thermal photons, they were self-trapped and turned into an excited state of coherent photons oscillating in phase with the quantum electromagnetic field (17). At this point, the Lamb-like shift, a critical number of coherent water molecules are tuned together and are enclosed within the CDs, where a phase transition occurs (17). The coherent oscillations of the molecules in the CDs no longer require any external supply of energy. It is stabilized and self-organized on all scale levels, which in the coherent state of FF-water from the Manawa vase and Greenhouse stack was identical to a strong power law relationship in the fractal scaling boundary of thermal IR emission flickering, characterized by high persistence, and the existence of CDs that enable a reservoir of quasi-free electrons (1,4). This observation revealed and agreed with the fact that liquid water can convert between two density states (6). At room temperature there are adjustable fluctuations between local water-molecular clusters of high-density (dominating around 80 %) and the low-density liquid.

The water CDs from the two high-grade energy producing FFs were activated to collect in their environment low grade energy, having a high entropy, and transform it, by exciting coherent vortices of almost free electrons, into high grade energy, with low entropy that can release this energy outward into useful work without thermal losses (18). Contrary, the activation of water by the Matatiki bowl and Glonn stack flow-path surfaces occurred without formation of a defined coherent state of water but on the other hand exhibiting specific high energy-grade chemical impact, which demonstrates differentiation in the high-grade energy state between the investigated FFs and their flowing waters, each specifically induced by its novel flow-path surface. The negative reductive charges may be single CDs or many nonlocal CD clusters. There may even be groups of rotating CDs moving in a loop producing local circulating interacting magnetic fields. Such circulating charges could appear over wide range of scales, from nanometers or less to many meters. The rotational speed of the reductive change depends on the resistance of the environment in which the charge moves. In special cases this resistance approaches zero, which equals a superconducting system mostly resulting from a quantum state (19). Our findings may support that the lower water TIRI temperature of the Manawa vase and Greenhouse stack and opposed to the non-activating FFs, tunes a transition water CD-state towards a higher dominance of local temperature density fluctuations, which characterizes the low-density liquid of water.

These CDs are receptors of weak non-local EMF signals (17). Water molecules become structurally clustered and attract not only adjacent water molecules but also other guest molecules able to resonate with same the frequency (18). With a further increase in density, the CDs become a net exporter of energy because the self-stabilized coherent state has a lower energy (12.06 eV), than the ionization threshold of an ordinary water state (12.60 eV) (17). Such coherent CDs contains millions of a low entropy state of non-local clustered water molecules, and a nonvanishing probability of having an infinite number of quasi-free electrons that can be donated readily to electron-acceptor systems. The quasi-free electrons, considered as the most significant property of a quantum coherent system, form frictionless vortices with an extremely long lifespan (15), lasting for days or weeks depending on a self-regulated magnetic moment that aligns coherent intrinsic and external vortices that cannot degrade thermally.

In view of the novel observation of a self-stabilized coherent FF water state from the Manawa vase and Greenhouse stack, the CDs could act as a donator of exited electrons or catalyze an excitation to a fairly-low activation energy to free them enough determined as a sustained reduction in the ORP in the bulk volume of FF-water, where the number of leaking electrons implies the tuned increase in pH. A concomitant array of water CDs whose extended coherence depends on the presence of non-aqueous guest molecules, endows the production of a magnetic vector potential from trapped EMFs of the water CDs (17,18), which was measured as a strong power law relationship in the fractal scaling boundary of the thermal IR emission flickering in ordered water, which is indicative of a high degree of self-similarity over multiple scales (20,21). Intriguingly, the production of a magnetic vector potential in the surrounding nonlocal space, whose rotor – and hence the magnetic field extends far in the environment (22). There exists, therefore a coupling between the vector potential produced by water CDs and the vector potential originating in the electrodynamics occurring in the adjacent or distant environment, like the electromagnetic radiation produced by sunspots, cosmic or atmospheric events (22), i.e. correlative to the *Bohm-Aharonov effect* (23).

The state of the underlying trend has a degree of autocorrelation, which is characterized by high persistence or long-term memory. The strong persistence in the DFA exponent indicates that the water temperature fluctuations from small temperature intervals to larger ones are positively correlated in a power-law fashion. Thus, a tendency towards decrease in water thermal IR emission is followed by another decrease in water IR emission at a different location related to fractal scaling behaviour. This implies that the long-range correlation should be considered an important factor in the trend prediction of water thermal IR emission. Additionally, the DFA exponent of water is strongly related to the capability of structural adaptation of the water body. Water with high DFA can maintain the fluctuation tendency of thermal radiation even when the external temperature changes, producing a decrease of entropy of the system and giving rise to potential self-organization (15).

Conclusion

A highly organized and coherent low-density liquid water state forms in the dynamic water flow of the designed flow-paths of the Manawa and Greenhouse stack Flowform vessels. The difference in vortical path of flowing water between low density activating and non-activating Flowform designs aligns with the fact that liquid water can convert between two density states. Despite variation in FF-water temperature fluctuations during study time the repeated formation of local low-density clusters was dominating in the dynamics of coherently flowing water. The results may support the quantum electrodynamics nature of the vortical flow-path surfaces of the Manawa vase and Greenhouse stack.

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Table 1. Thermal infrared (IR) emission imaging, fractal scaling and water chemistry in spring water circulating in flowforms for two hours.

Flowform	Water Temperature		Fractal scaling			Water chemistry		
Sample[a]	Temperature (ºC)	Temp Diff (ºC)	TIRI Temp (ºC)[b]	TIRI Temp Diff (ºC)[b]	DFA[c]	pH[d]	ORP (mV)[d]	Conductivity (µS)[d]
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[a]Water strored in polystyrene cells. All values are an average of 20 water samples analyzed occationaly for 2 months on each Flowform.

[b]TIRI Temperature was calculated on numerical pixel data, each sample calculated on six repetitive image frames taken in sequence.

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